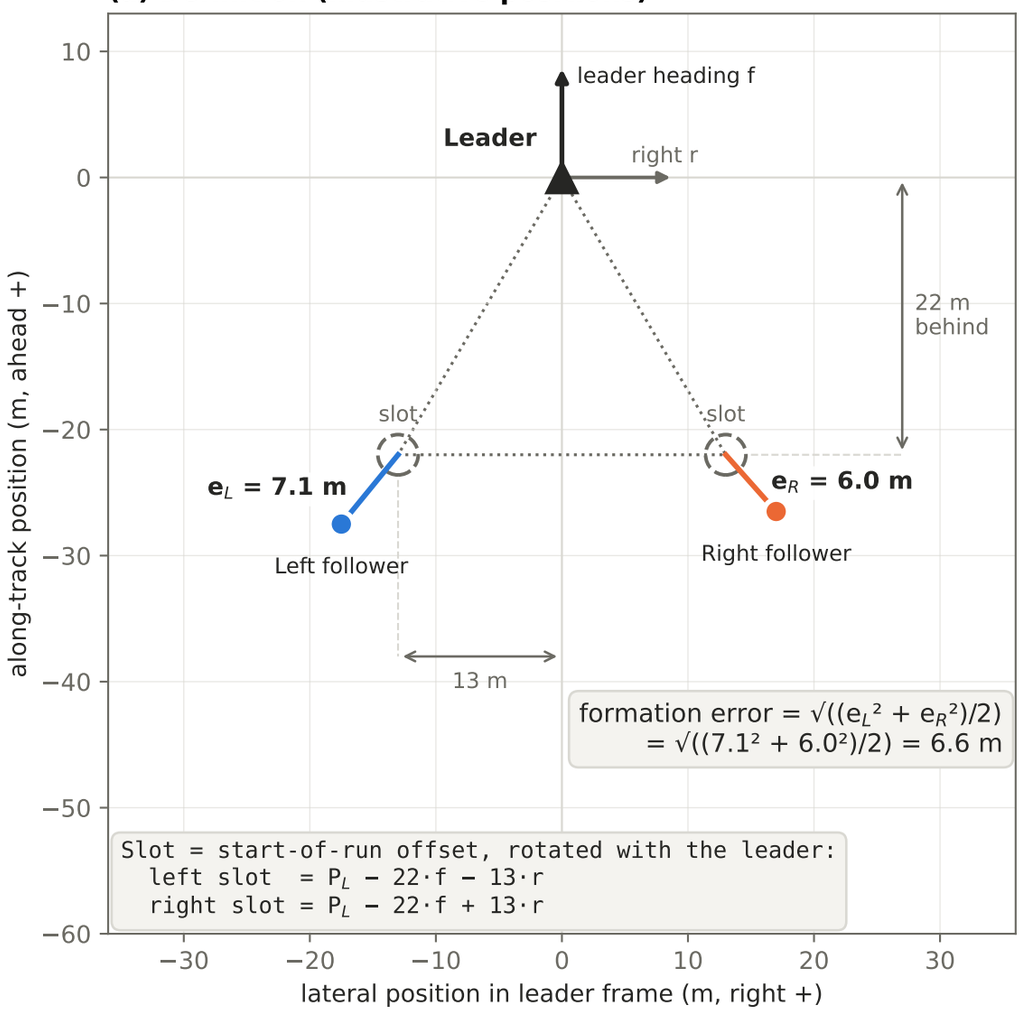
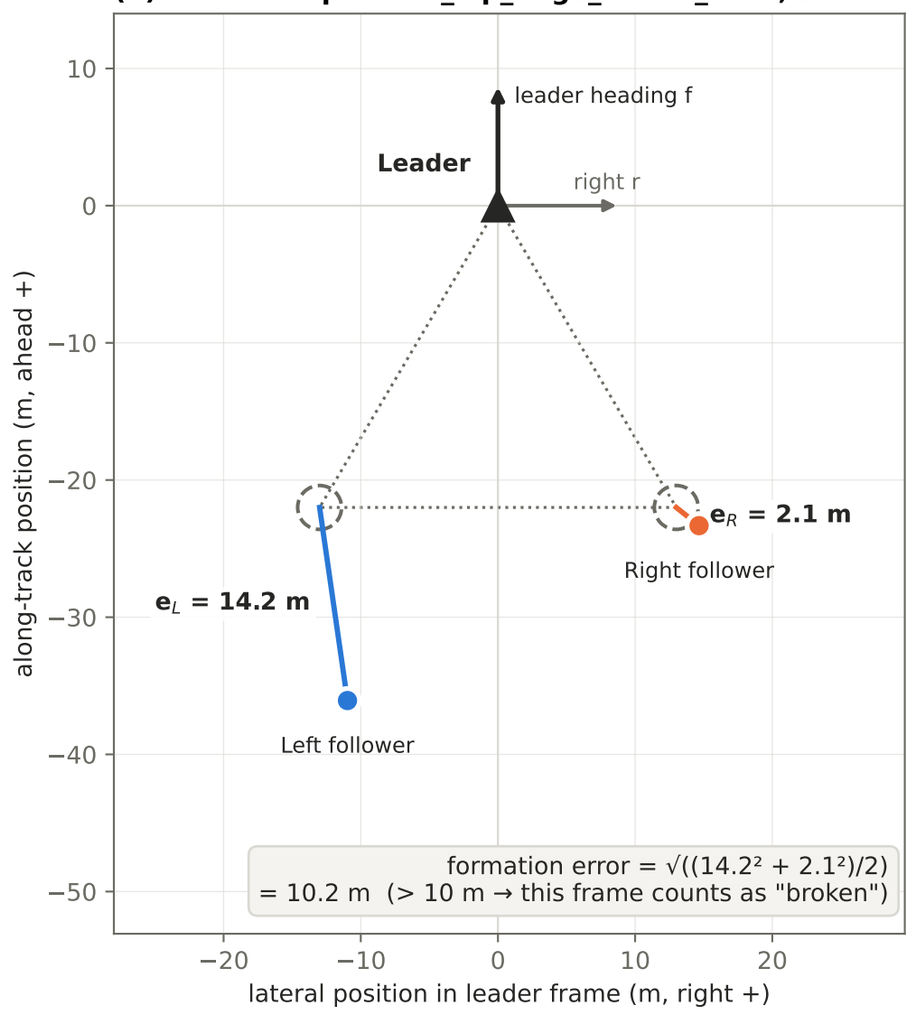
How formation error is calculated (rollout videos and experiment matrix)

Positions come from simulator ground truth (evaluation only — never given to the boats). Errors are horizontal distances in metres.

(a) Definition (illustrative positions)



(b) Real example: L03_lap_huge_waves_sea3, seed 4903, Kalman OFF, t = 113.8 s



(c) Over a run: per-frame formation error → Final RMSE and broken %

